Infrastructure University Kuala Lumpur

IUKL . Engineering Your Future

HUNTER'S PET APP

Background Studies and Outline

Introduction

With the development of AR (Augmented Reality) technology, more and more people start to use AR technique for a software or application for enhancing user experience. AR technology is the combination of virtual objects and reality. It increases user experience on software and application. Nowadays, people's living standards and emotional needs keep increasing. Keeping pets has become a popular trend to their daily life. Many people would like to keep a pet to accompany them. However, some people do not have the opportunity to have their own pet due to several reasons even they really love and enthusiastically desire to have a pet. In order to let them having their own pet for companionship, "Hunter's Pet App" is an Augmented Reality (AR) mobile game that develop for everyone to have a virtual pet in reality. Through this mobile AR game, the user may have their own virtual pet and may not be concerned about their ability or economy to take care of it. It allows users to keep their pet in an entertaining way.

Problem Statement

1. Pet lover cannot have their own real pet
Some people really love to have a pet as their pet companion. However, they might not be able to keep and nurturing a real pet due to several affordability and limitation. For example, some people cannot have it due to allergies, family issues, economic concern or their living environment did not have space for the real pet. However, a virtual pet no need to take physical space. Therefore, there may be certain reasons and conditions that cause them cannot keep a real pet.

2. First-time dog owners confused what kind of dog breed is suitable for them
Some pet lovers get confused before they keep and nurturing a real pet. There is much kind of dog breeds in our world. Each dog breed has different characteristics, personalities and needs. First-time dog owners may not know what type of dog breed they want due to lack of knowledge on dog breed. Thus, the proposed system enables them to acquire knowledge about dog breeds and provides them with experience in caring for a dog breed they really like..

3. Increase amount of abandoned pet due to lack of responsibility and patient on real pet
Basket drop-off is a project managed by SSPCA which allow people to make donation in the form of donation-in-kind. Donor can donate the prime necessities needed by the shelter such as pet food, cleaning aids or other necessities such as animal leashes, shampoo, etc. Currently, there are multiple basket drop-off points at various pet shops available in Kuching and Samarahan. But, there is difficulty in letting enable donor to drop-off the donation since there is insufficient information available for them to refer to.

4. Most of the existing virtual pet of the virtual pet game does not look realistic
A lot of virtual pet games have been developed today. However, most of the existing virtual pet game is not look real and attractive enough. Even the virtual pet game were developed with attractive graphics, animation and good features, the virtual pet is still only can show up and appear on people's device screen. The virtual pet game will not exist or appear in the real world environment. This situation may gradually make people feel bored and lost interest on the game thereby abandon their virtual pet of the game. Therefore, an Augmented Reality (AR) technique is needed for the proposed system.

Objectives

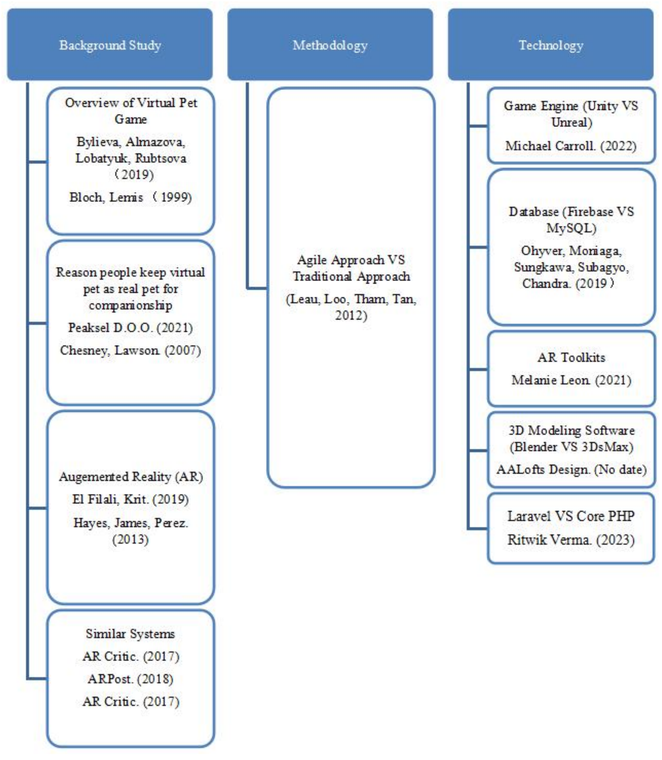
1. To develop a mobile AR game that provides opportunity for user to choose the virtual pet dog breed based on their preferences.
2. To allow individual to accompany and view their pet dog to grow up.
3. To allow users nurturing their AR Pet through feeding and playing.
4. To allow user to earn rewarded or coins.
5. To develop a virtual pet shop mechanism to let users purchase and gain the pet's needs.
6. To enable users give training command to pet dog.
7. To allow people entering a virtual environment to view and explore information that related with pet dog for raising people's knowledge and awareness about dog.
8. To develop a website that allow administrator to manage the game's data.

Scope of Work

Administrator Scope	Player Scope	System Scope
• Able to login the game management system. • Able to do CRUD (create, read, update, delete) management on player data.	• Able to register new account. • Able to login to the game. • Able to modify account data. • Need to select and give name to the virtual pet dog they like. • Scan the surrounding environment for placing their virtual pet dog. • Able to nurturing the pet dog with feeding, watering and playing. • Able to provide training command to their virtual pet dog. • Able to earn rewarded or coins for gaining pet's physical needs such as dog food and dog treat. • Able to shop and purchase the product for virtual pet dog. • Able to view the pet's health and growing status. • Able to view and explore about dog information in a virtual world by entering a virtual portal that placed in real world.	• Able to do plane or surface detection for placing the virtual pet dog. • Able to display the virtual pet dog in real environment with AR (Augmented Reality) technique. • Able to store and display the game data. • Able to display animated virtual pet dog. • Display virtual shop for player to buy product for their virtual pet dog.

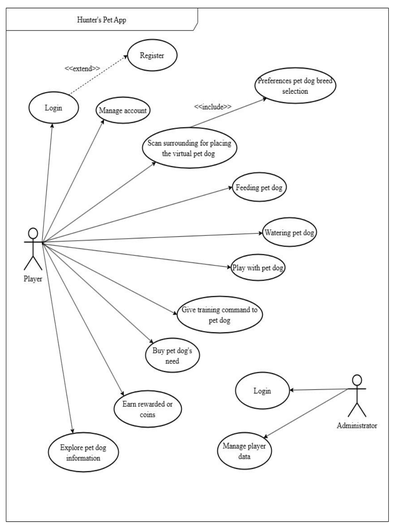
Literature Review

Literature Review Map/ Summary Table

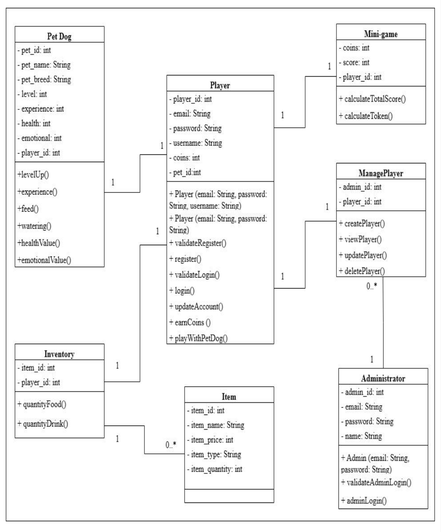


Methodology

Functional Model

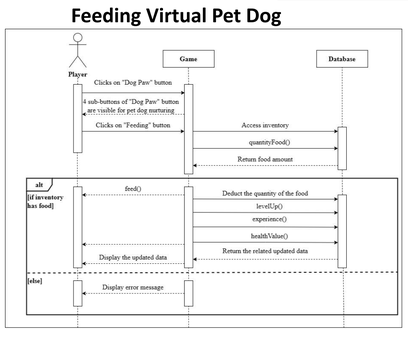


Structure Model

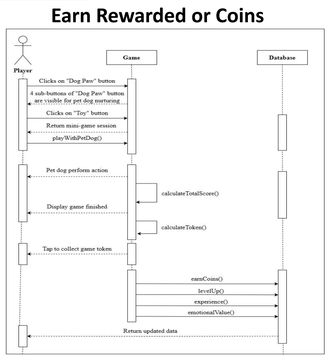


Behavioural Model

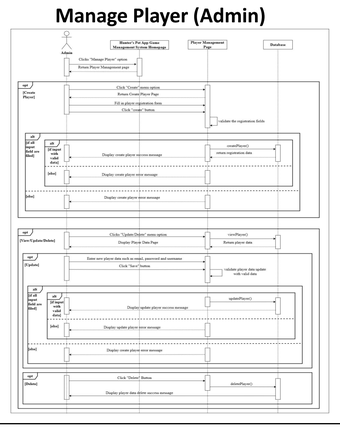
Feeding Virtual Pet Dog



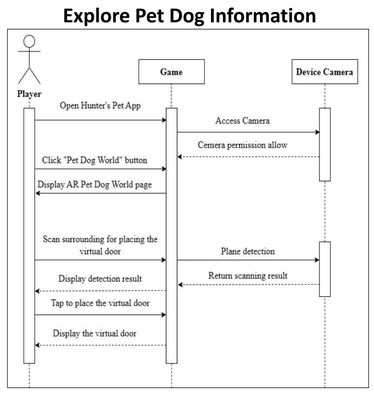
Earn Rewarded or Coins



Manage Player (Admin)

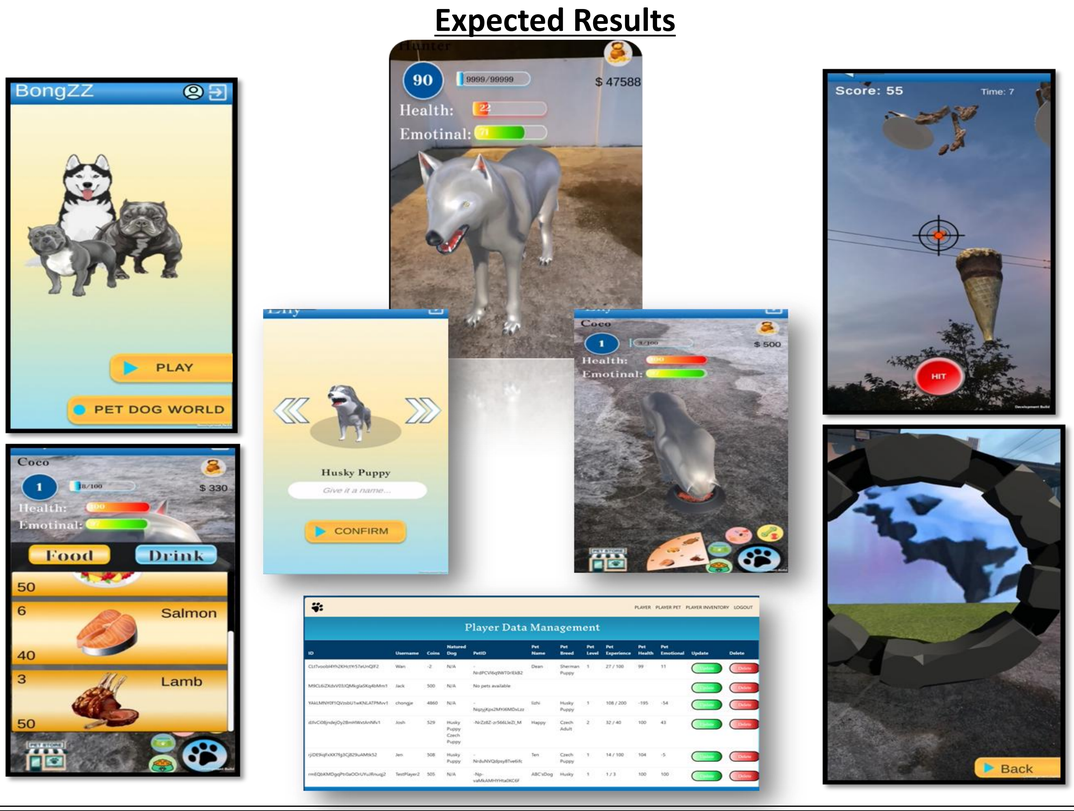


Explore Pet Dog Information



Expected Results and Contributions

Expected Results



Conclusion

In conclusion, Hunter Pet App is providing users the chance to keep their own pet dog and gaining experience on how to take care of a pet dog in an entertaining way. Therefore, it is necessary to develop the virtual pet game for people. The project aims to solve the problem that people cannot keep real pets. It may help people to keep a pet without any stress and concern. In addition, the project with AR technique allows the virtual pet become more realistic and engaging people interest to play with it.

References

1. Bloch, L. R., & Lemish, D. (1999). Disposable love: The rise and fall of a virtual pet. New Media & Society, 1(3), 283-303.
2. Bylieva, D., Almazova, N., Lobatyuk, V., & Rubtsova, A. (2019, October). Virtual pet: trends of development. In The 2018 International Conference on Digital Science (pp. 545-554). Cham: Springer International Publishing.
3. Chesney, T., & Lawson, S. (2007). The illusion of love: Does a virtual pet provide the same companionship as a real one?. Interaction Studies, 8(2), 337-342.
4. El Filali, Y., & Krit, S. D. (2019). Augmented reality types and popular use cases. International Journal of Engineering, Science and Mathematics, 8(4), 91-97.

Name: **BONG LI ZHI**

ID: -----

Program Code: -----